Milestone 1 — Data Exploration & Baseline Analysis

Smart MCQ Solver Challenge · Introduction to DL & GenAI [BSDA2001P]

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> **Objective.** Understand the dataset, establish simple non-learned baselines, and verify the evaluation metric (MAP@3) before building any trained model. The central question of this milestone: *how far can naive, lexical methods get us — and what do they reveal about the task?*

1. Dataset Overview

The training set contains **2,000 questions** across 8 columns: an `id`, a `prompt`, five answer options `A–E`, and the correct `answer` label. Each question has exactly one correct option, and the task is to rank the options and submit the top three (metric: MAP@3).

2. Exploratory Data Analysis

2.1 Answer distribution

The correct-answer labels are mildly imbalanced — option B is most common, option E least:

Answer	Count	Share
B	490	24.5%
C	459	23.0%
A	369	18.5%
D	358	17.9%
E	324	16.2%

No single class dominates, so a model cannot succeed by always predicting one letter. However, the imbalance is real and is worth correcting later with balanced class weights.

2.2 Vocabulary and text statistics

- Unique words across all cleaned prompts: **859** (lowercased, punctuation removed).
- TF-IDF vocabulary over combined prompt + all five options: **2,762** terms.
- A sample prompt of 22 words reduced to **13 content words** after English stop-word removal — confirming that a large fraction of prompt text is boilerplate ("pick the best possible answer", "among the listed options", etc.) carrying no answer signal.

3. Metric Verification — MAP@3

The competition metric, **Mean Average Precision at 3**, was implemented and unit-tested. For a single question it awards `1/rank` if the correct answer appears in the top-three predictions:

True answer	Ranked prediction	MAP@3

C	[C, A, B]	1.0000
B	[D, B, E]	0.5000

Correct at rank 1 scores 1.0, at rank 2 scores 0.5, at rank 3 scores 0.33, otherwise 0. The implementation matches these expected values.

4. Baseline Experiments

4.1 Majority-class (frequency) baseline

Predicting the three globally most frequent answers `[B, C, A]` for *every* question — a model that ignores the text entirely — yields:

> **Frequency baseline MAP@3 = 0.4213**

4.2 TF-IDF cosine-similarity baseline

A slightly "smarter" lexical approach: vectorise the prompt and each option with TF-IDF, then rank options by cosine similarity to the prompt. The intuition is that the correct answer should share vocabulary with the question.

- Cosine similarity, prompt vs. option A of question 1: **0.2720**.
- Top-1 accuracy across all 2,000 questions: **13.55%** (271 correct).
- Ranking MAP@3: **0.2962**.

5. Key Finding

> The text-based TF-IDF cosine method (**MAP@3 = 0.2962**, accuracy 13.55%) performs **worse** than the text-blind frequency baseline (**MAP@3 = 0.4213**) — and worse than simply always guessing B (24.5%).

This is the most important result of the milestone. It shows that in this dataset the correct answer is **not** the option most lexically similar to the prompt; surface word-overlap is actively misleading. The questions require semantic understanding, not keyword matching. A naive similarity model is therefore a weak baseline, and any real gain must come from a model that learns meaning — which motivates the trained models built in Milestone 2 (a from-scratch BiLSTM, a tuned TF-IDF + Logistic Regression classifier, and a fine-tuned DeBERTa transformer).

6. Summary of Results

Method	Uses text?	MAP@3
Frequency baseline (predict B, C, A)	No	0.4213
TF-IDF cosine similarity ranking	Yes (lexical)	0.2962

7. Conclusion & Next Steps

- The dataset is clean, mildly class-imbalanced, and small (2,000 questions).
- Prompt text carries significant boilerplate; stop-word / instruction stripping will help.

- Lexical similarity is insufficient — the task needs learned semantic representations.
- **Milestone 2:** build and compare three learned models, and — critically — design a *leakage-free* validation split, since early inspection suggests duplicate questions exist in the data.